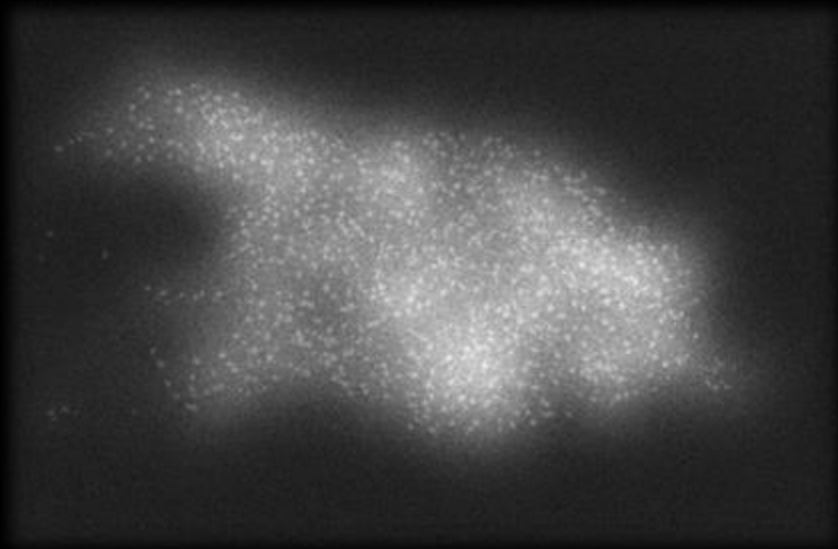




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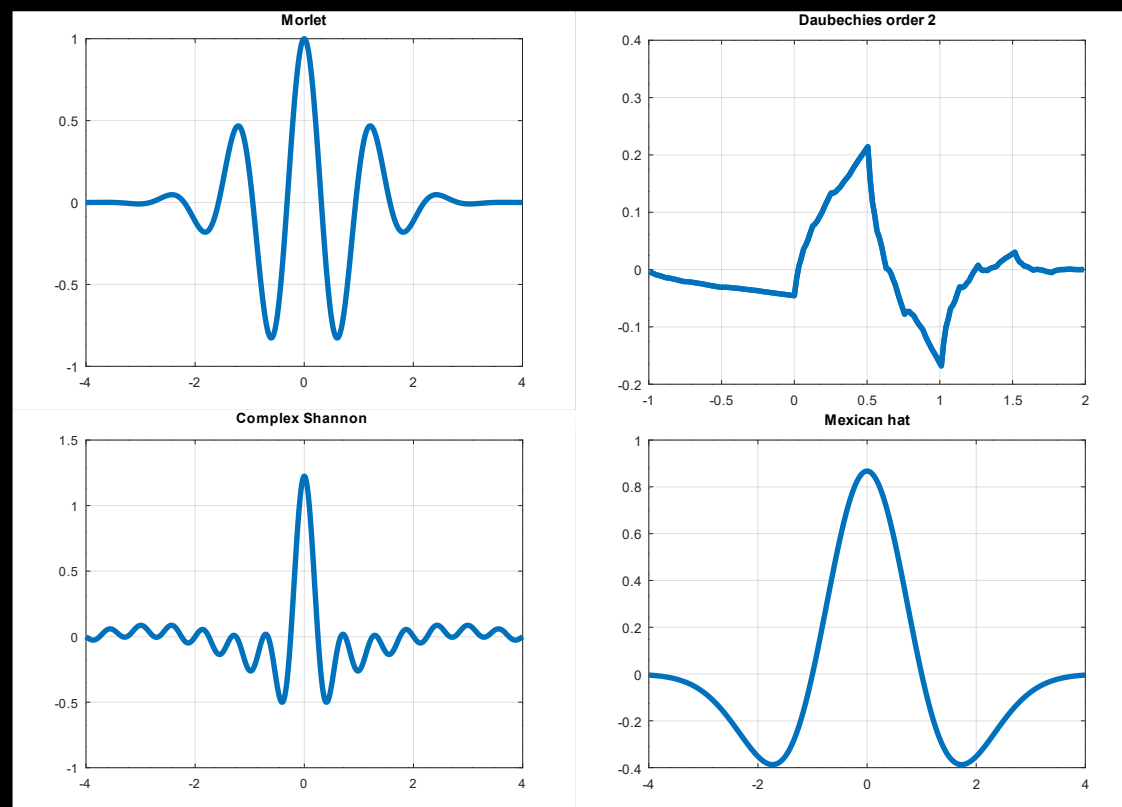
THE LEARNING WAVE

ADAPTIVE THRESHOLDING FOR BRIGHT SPOTS DETECTION IN NOISY IMAGES

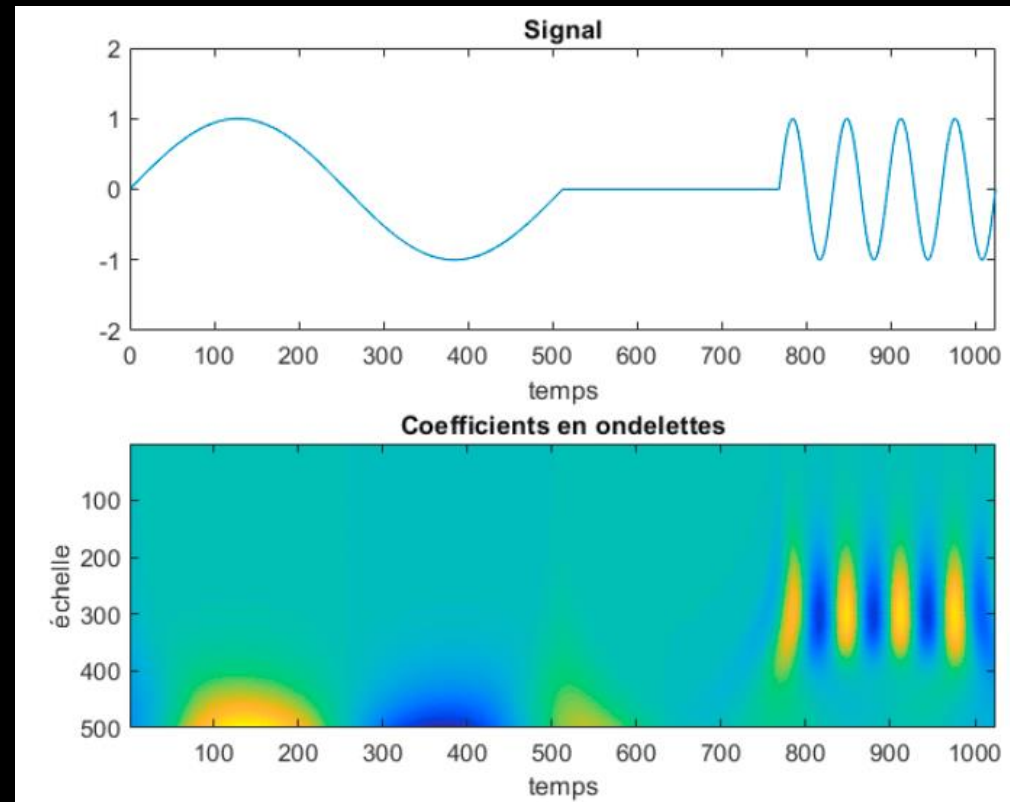
Lab meeting – August 26th, 2025

Corentin Nicodème

WHAT ARE WAVELETS?

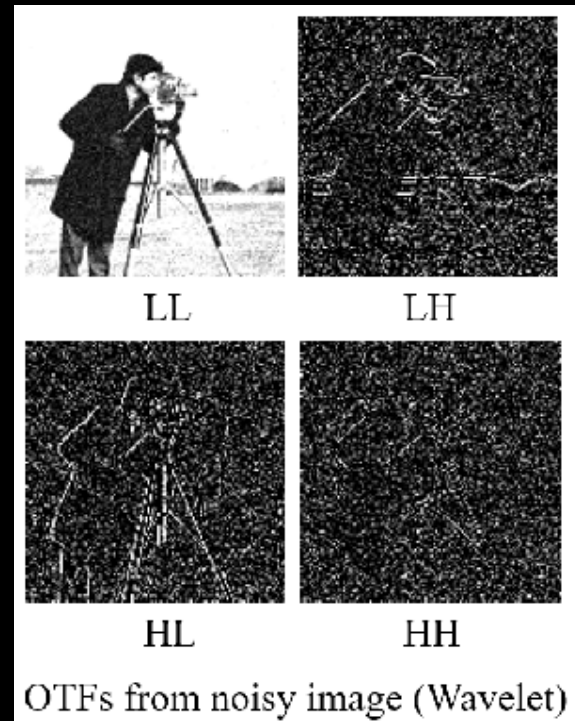


WHAT'S IT FOR?



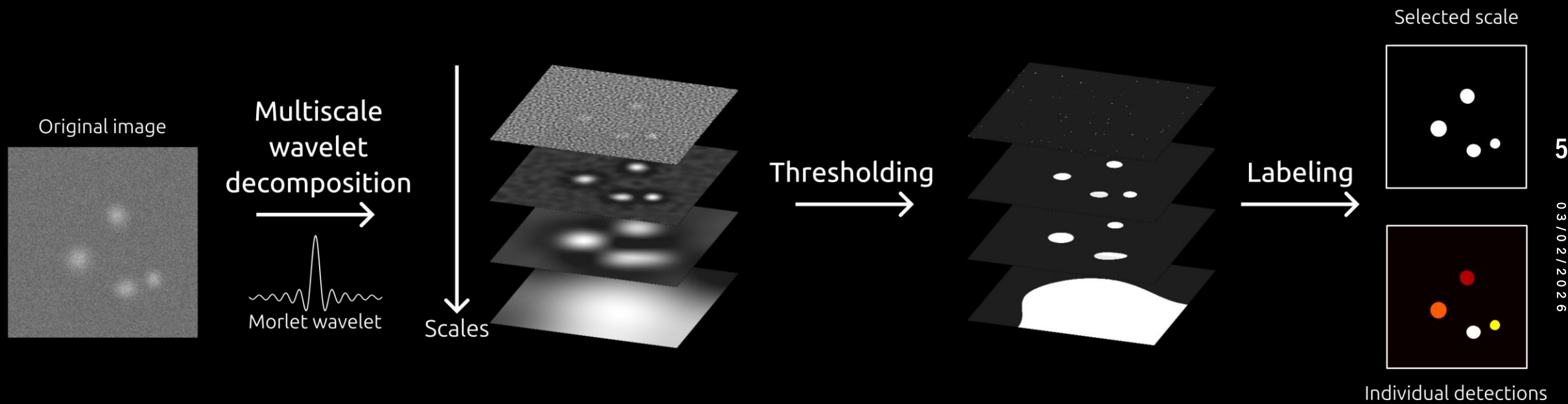
Wavelet transform

WHAT ABOUT THE IMAGES?

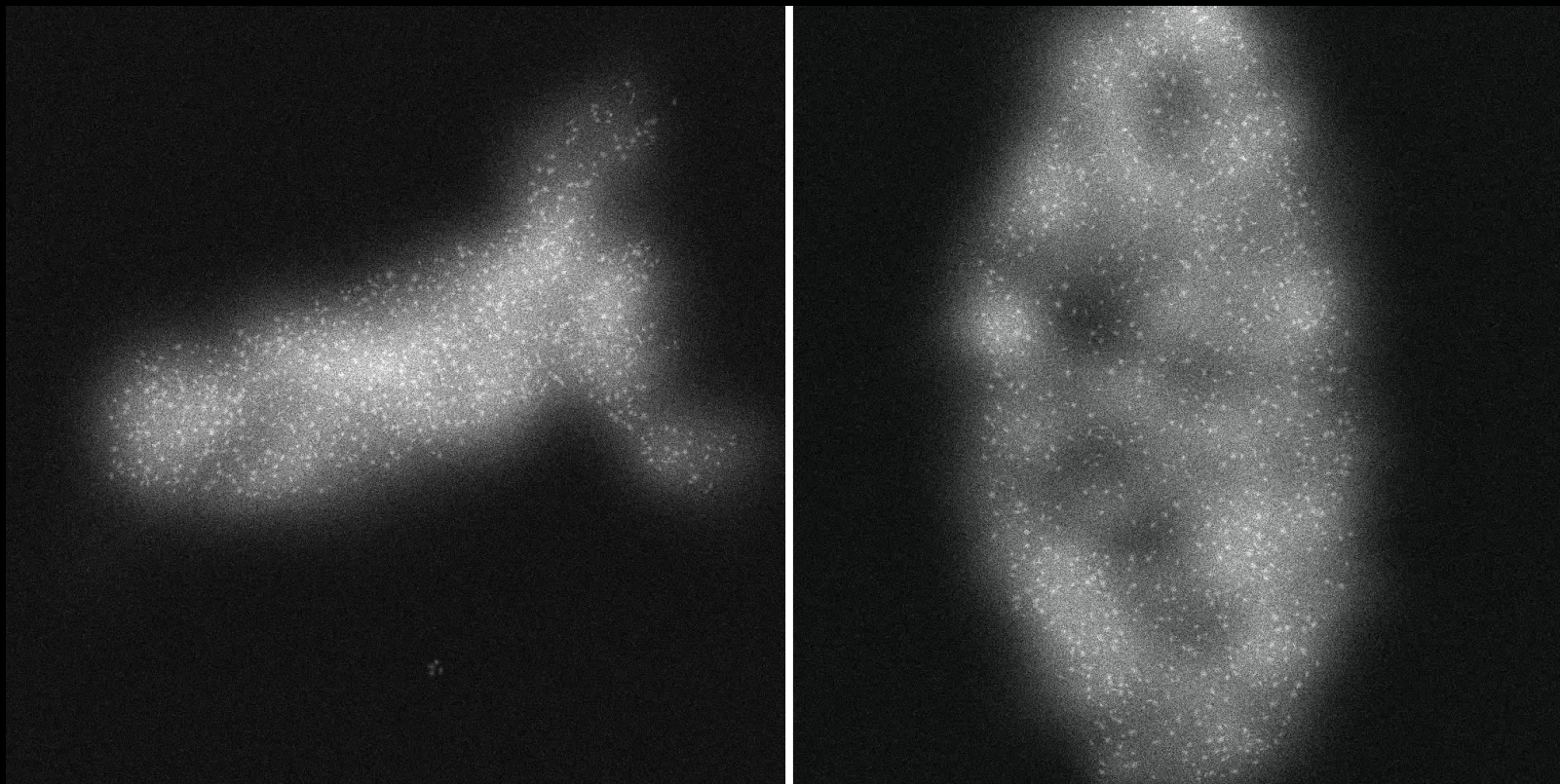


OTF : orthogonal
transform features

WHAT ABOUT THE IMAGES?

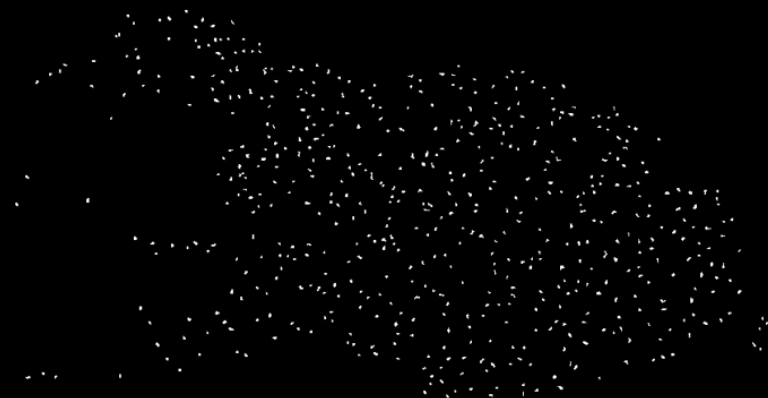
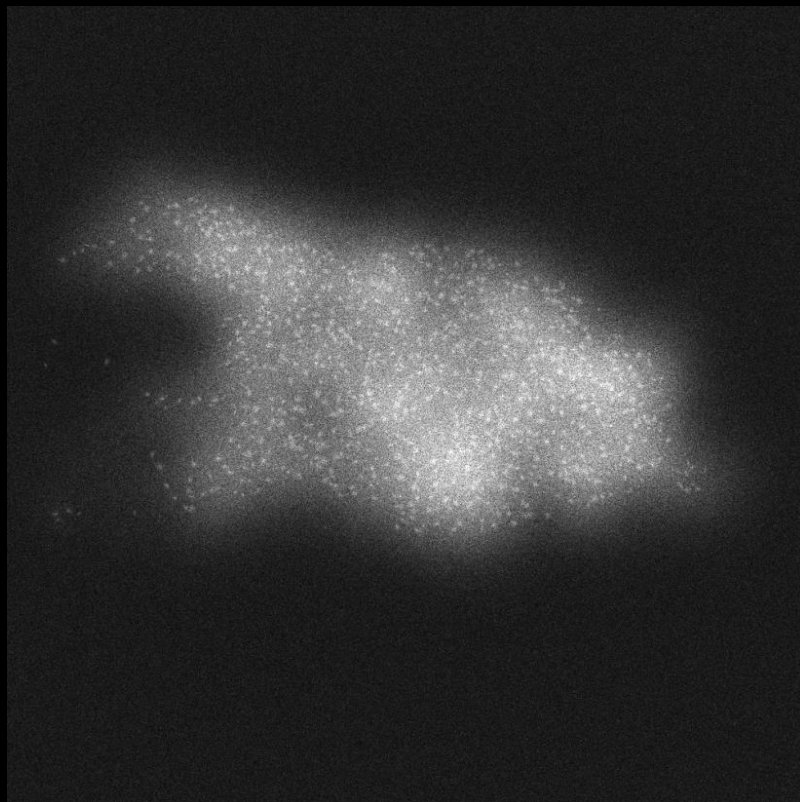


FIRST APPROACH: THE SIMULATOR

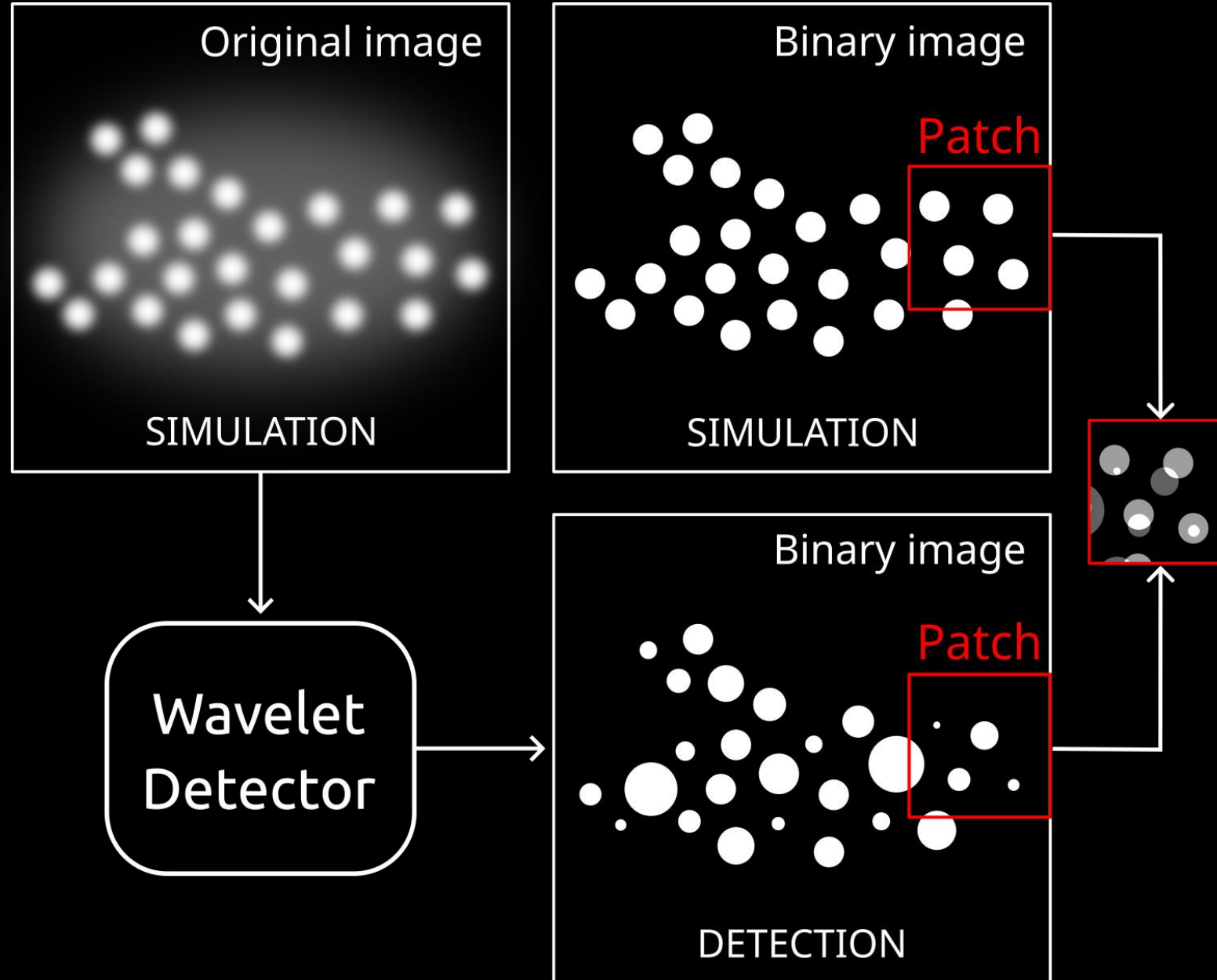


SINETRA simulator

RESULTS ON SIMULATED IMAGES

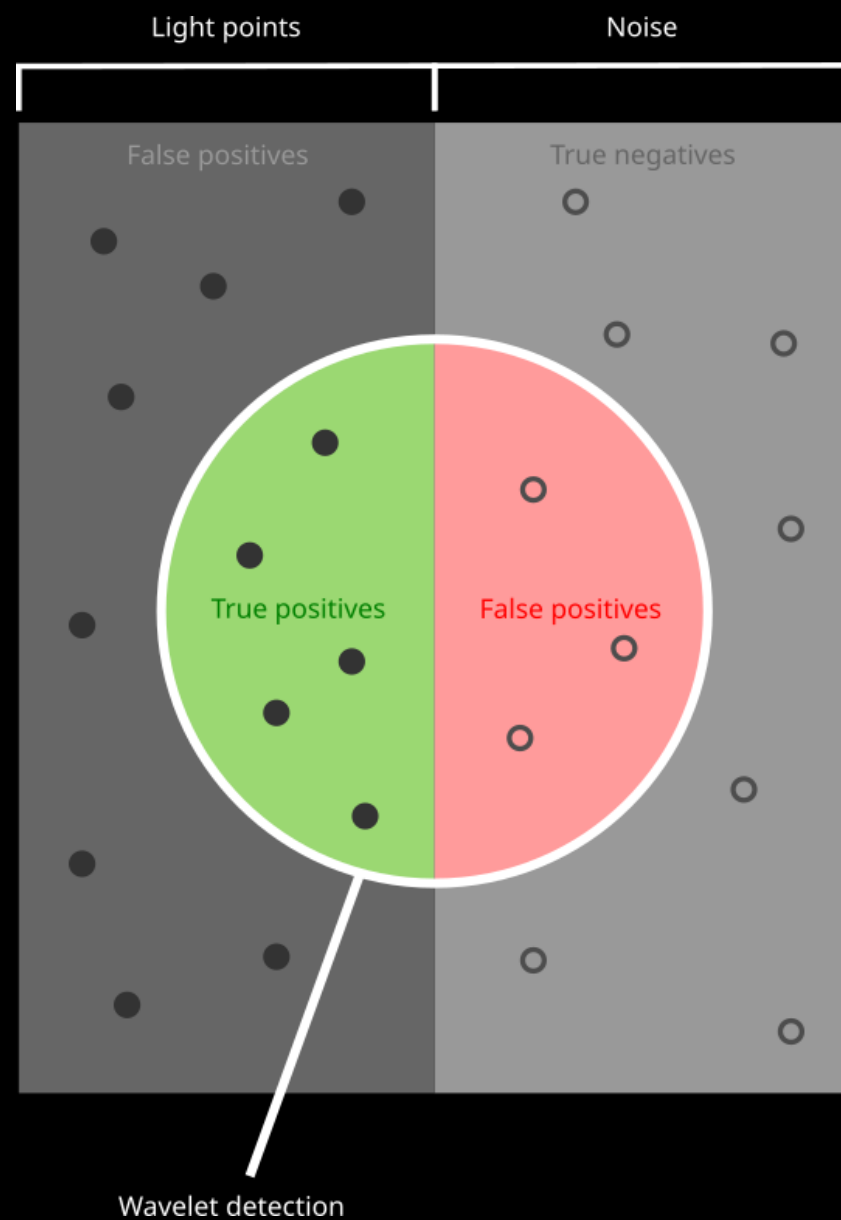


METRICS



METRICS

F-score

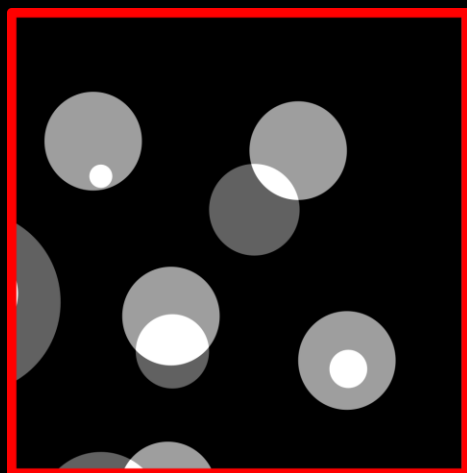


$$\text{Precision} = \frac{\text{True positives}}{\text{True positives} + \text{False positives}}$$

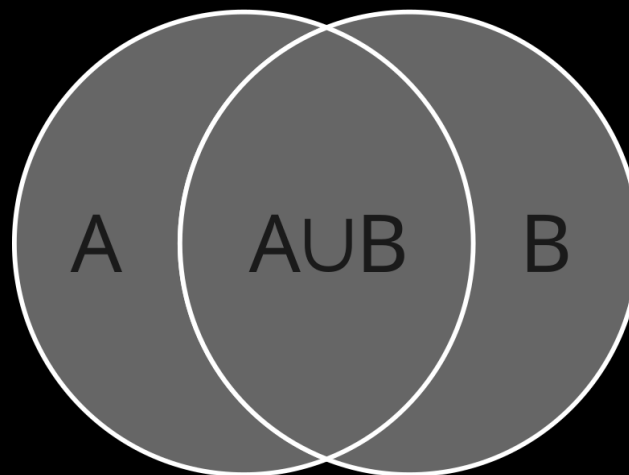
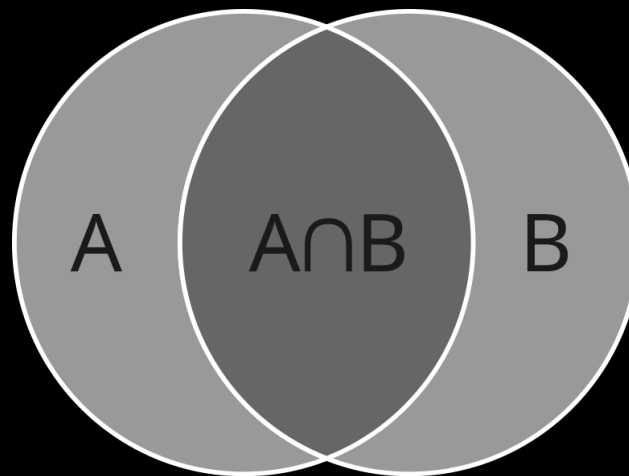
$$\text{Recall} = \frac{\text{True positives}}{\text{True positives} + \text{False negatives}}$$

$$F_1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

METRICS

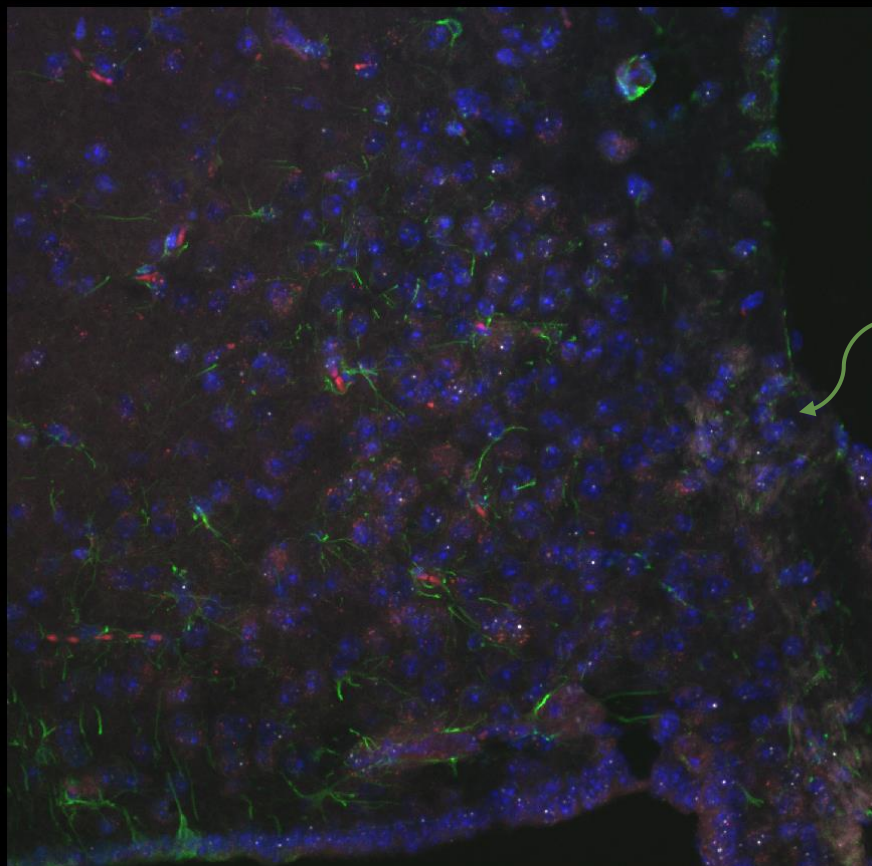


Jaccard index



$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

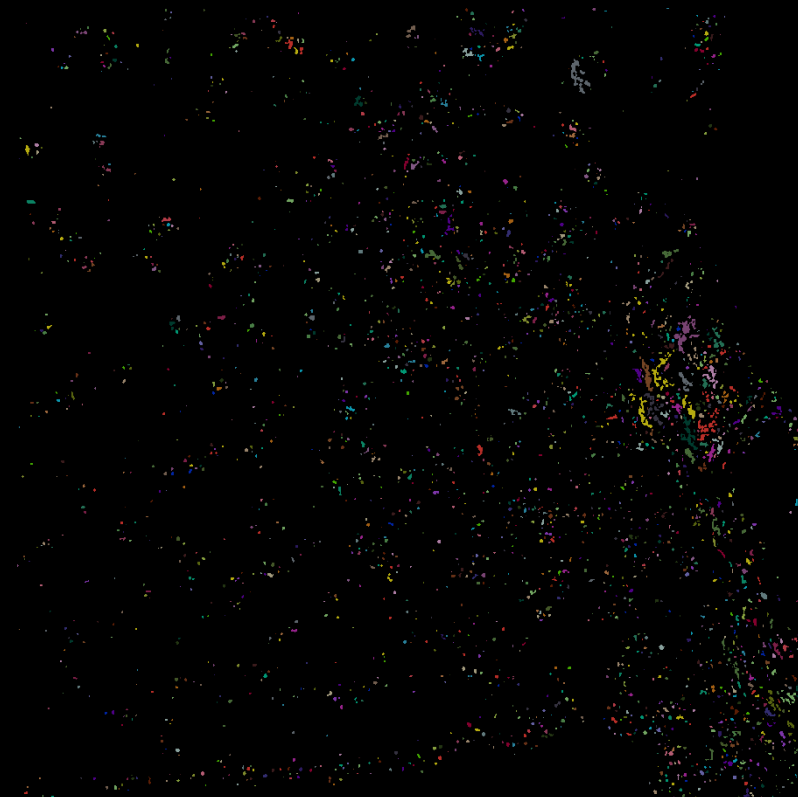
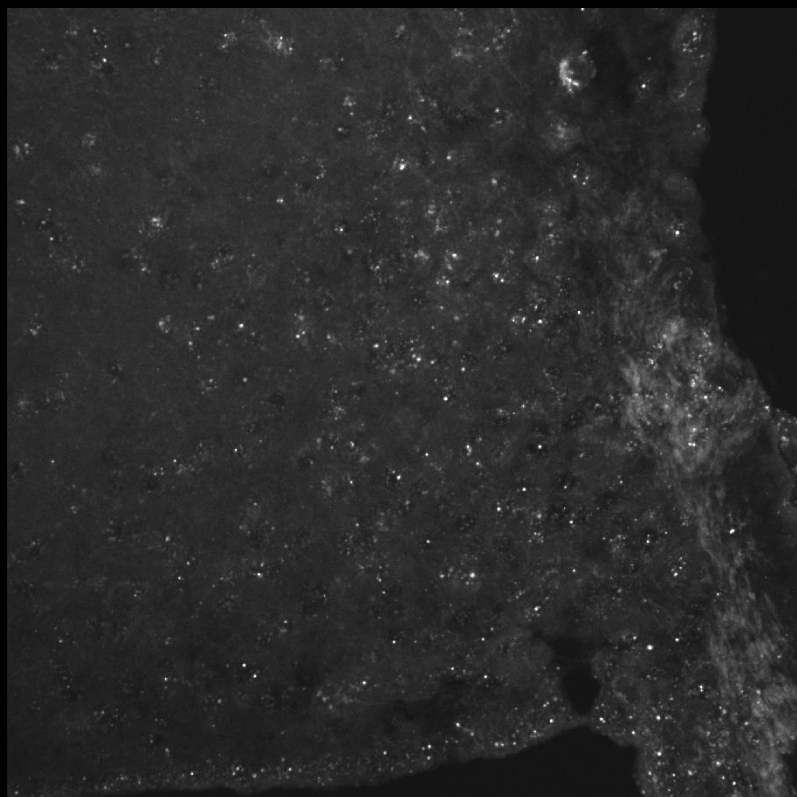
AND FOR REAL IMAGES?



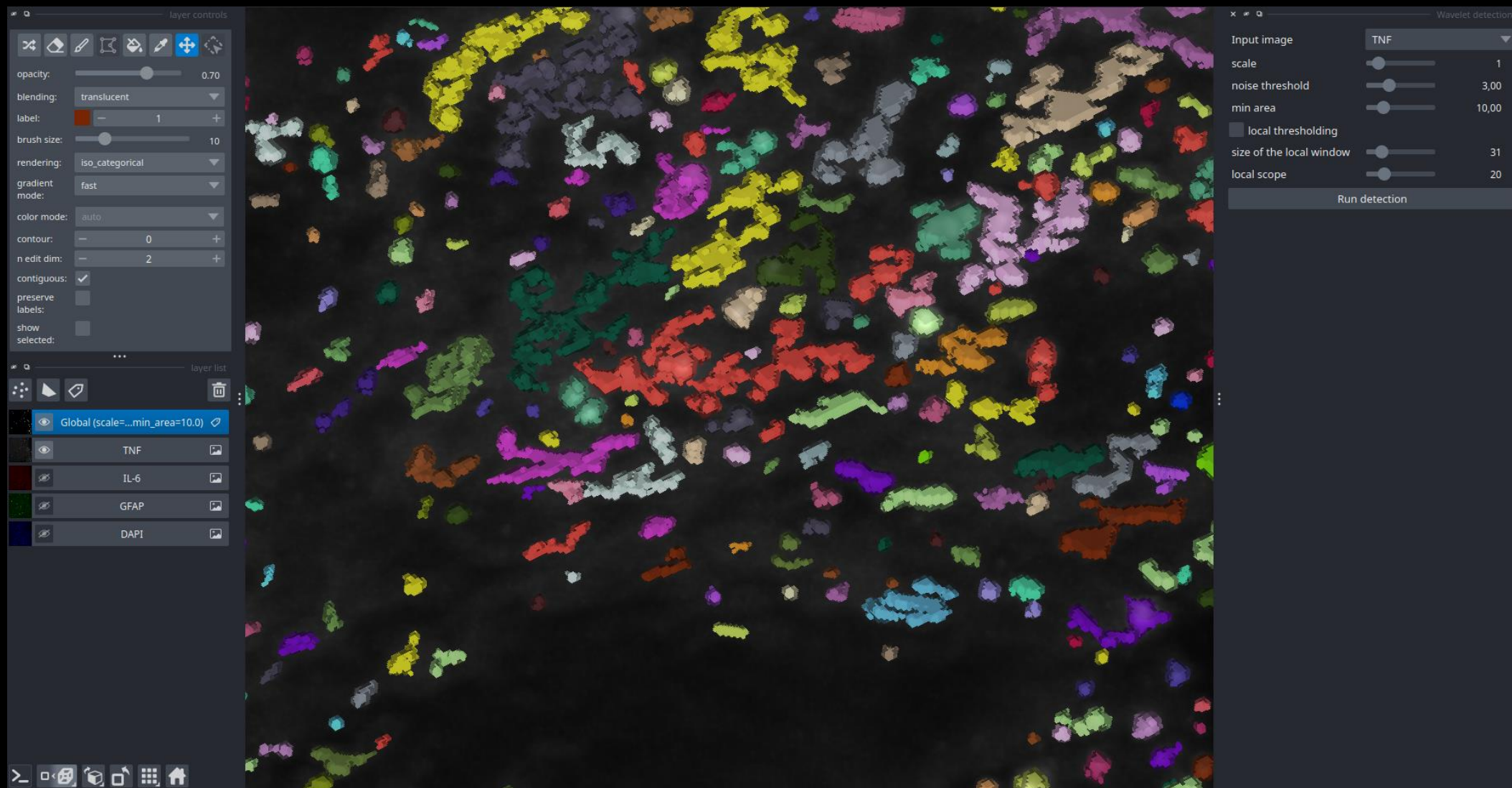
Brain cells
(astrocytes)



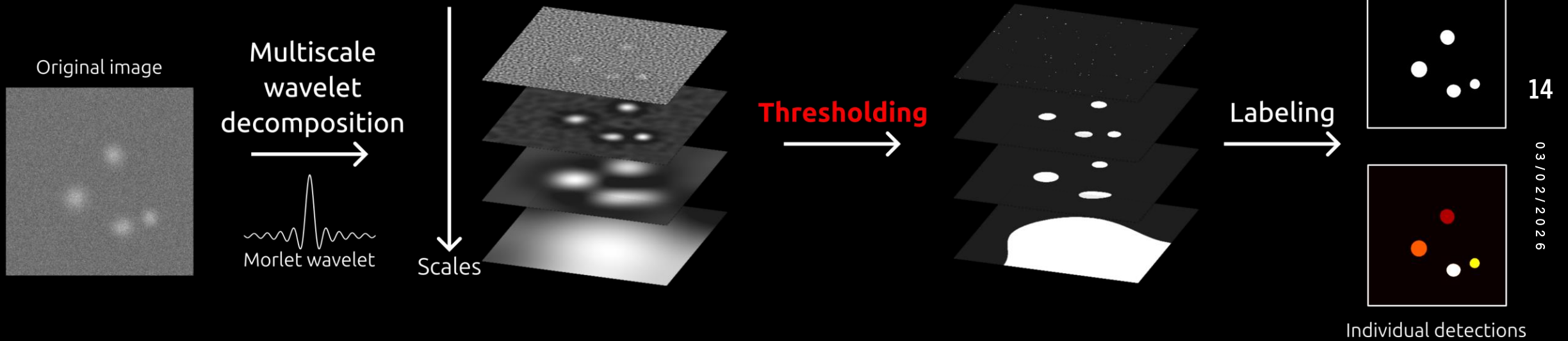
RESULTS ON NATURAL IMAGES



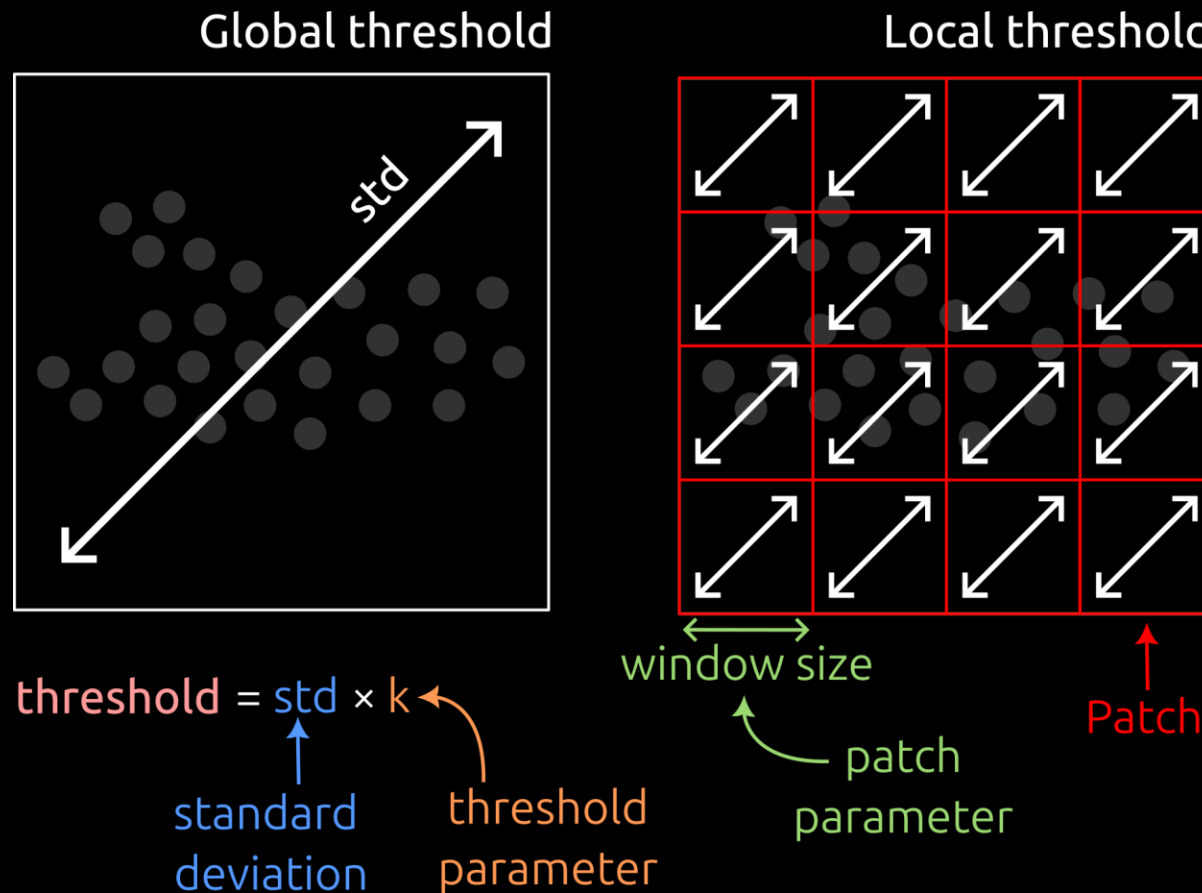
RESULTS ON NATURAL IMAGES



WHERE CAN THE DETECTOR BE IMPROVED?



GLOBAL VS LOCAL THRESHOLD



$$\text{local threshold} = \text{clamp}(\text{std} \times k, \text{min}, \text{max})$$

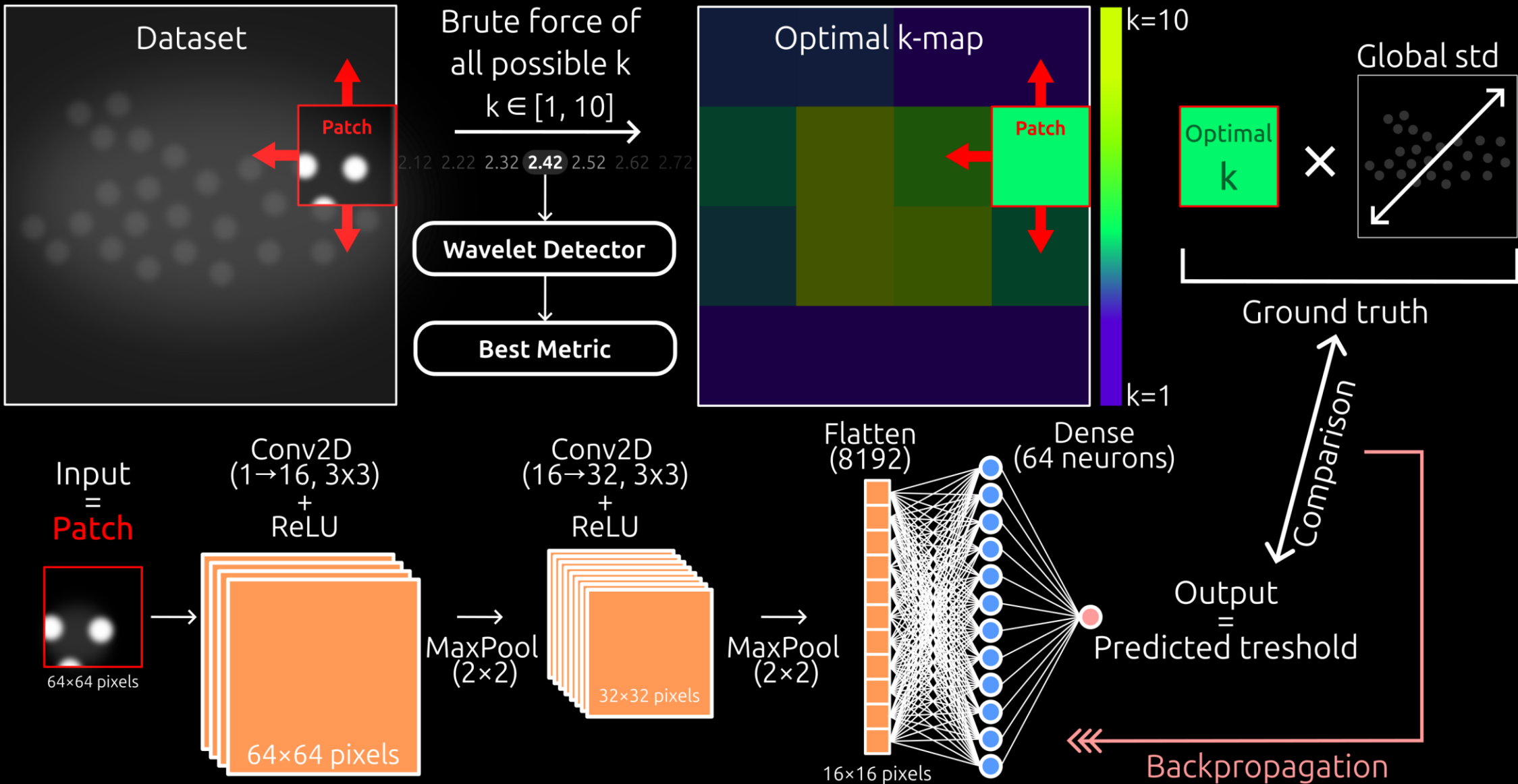
$$\text{min} = \text{global threshold} \times (1 - \text{local scope})$$

$$\text{max} = \text{global threshold} \times (1 + \text{local scope})$$

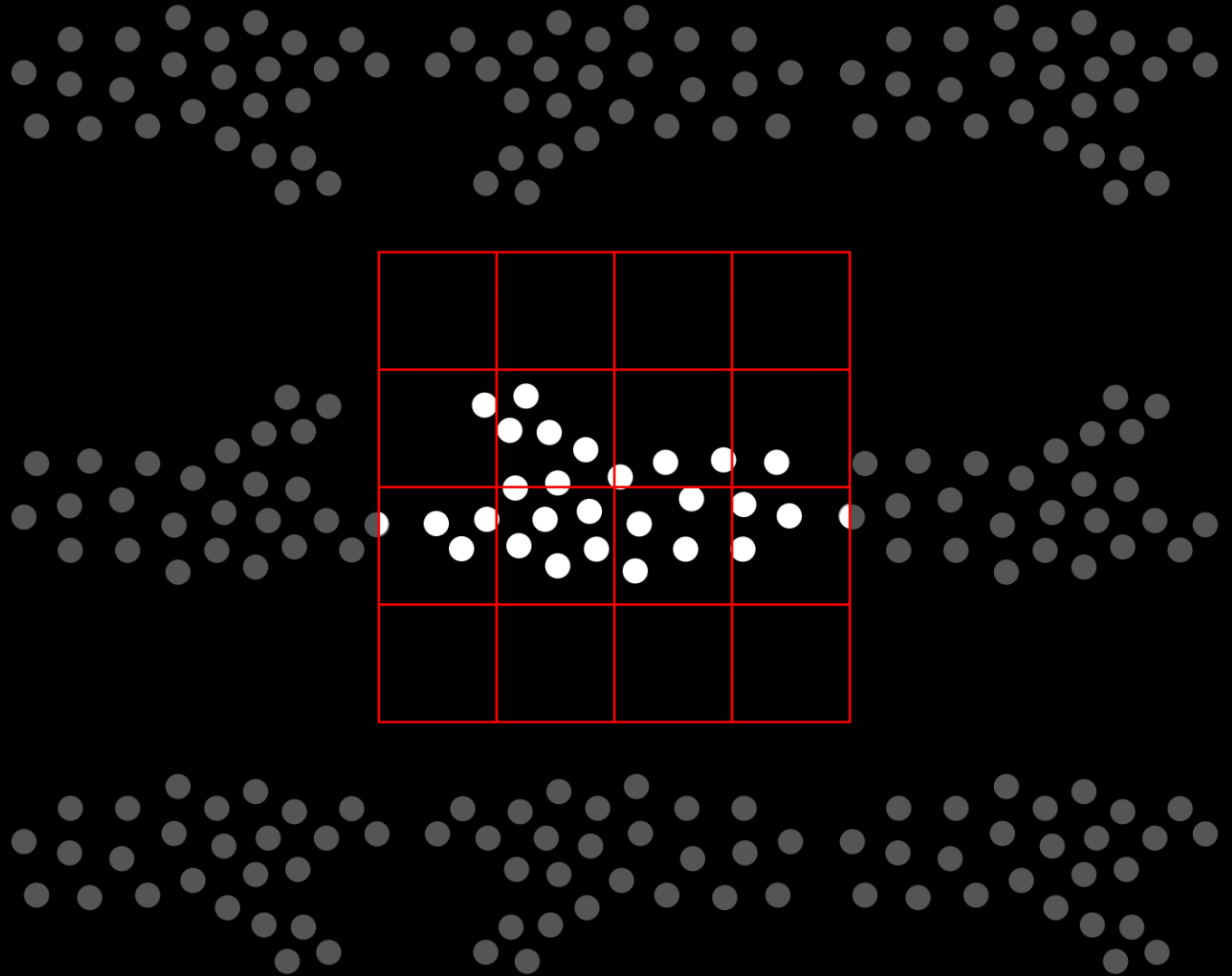
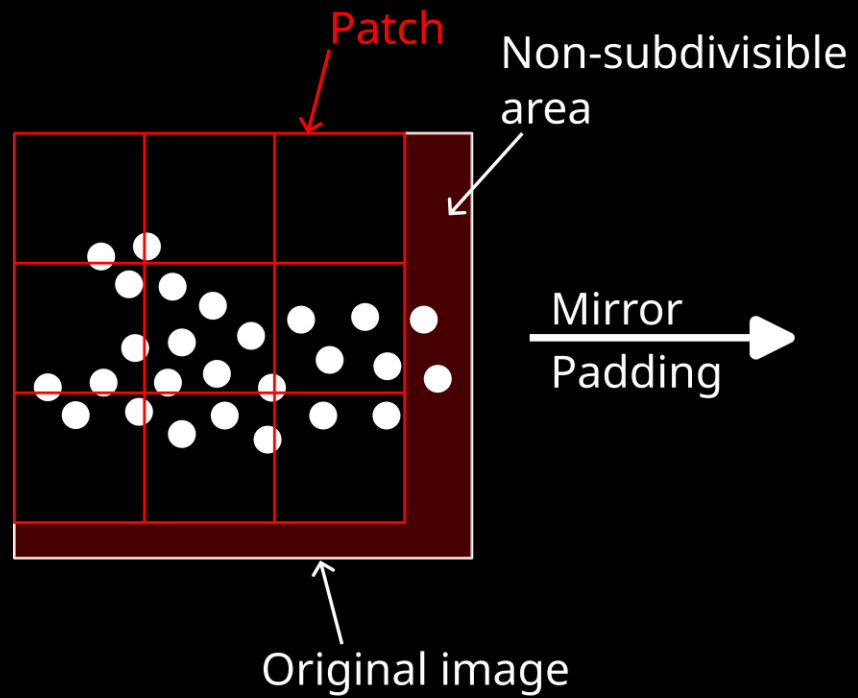
$$\text{local scope} = 0.2$$

WHAT ABOUT AI?

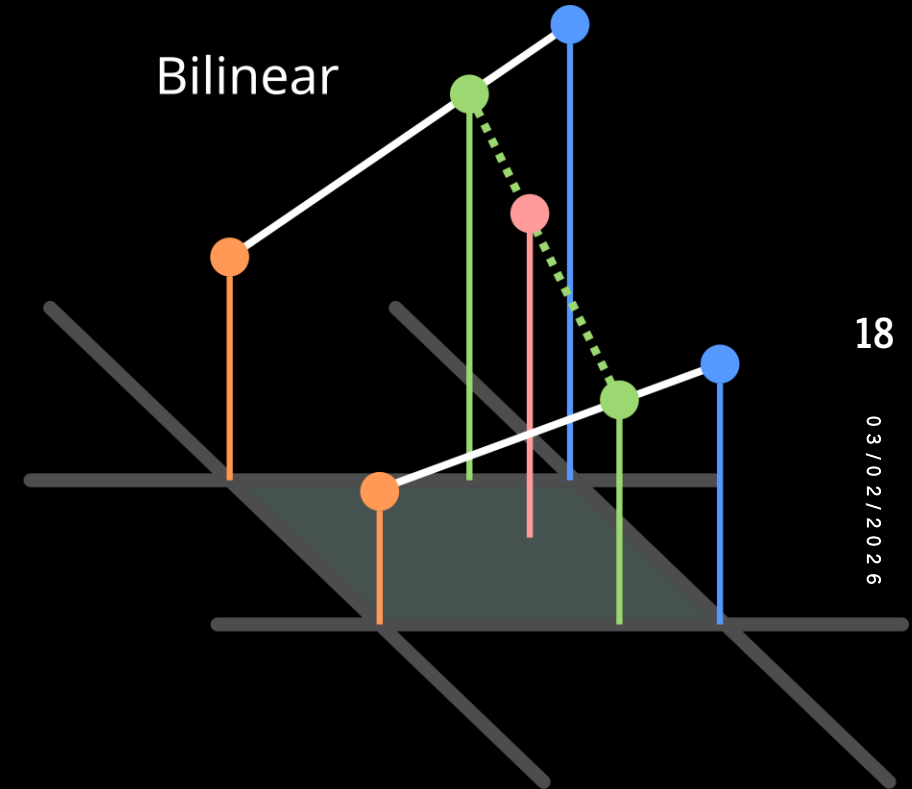
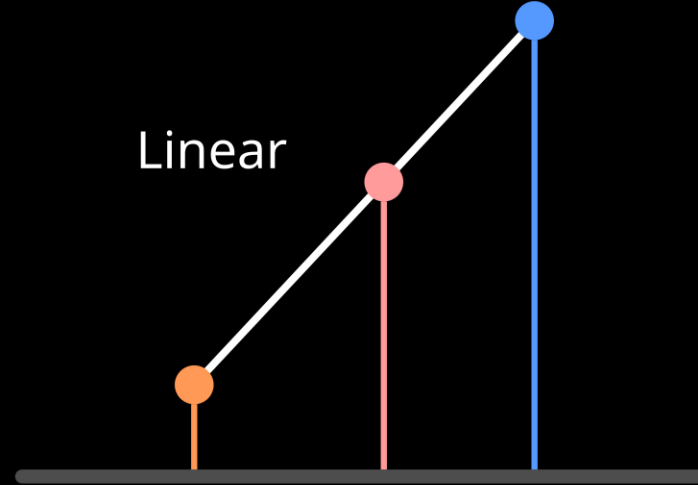
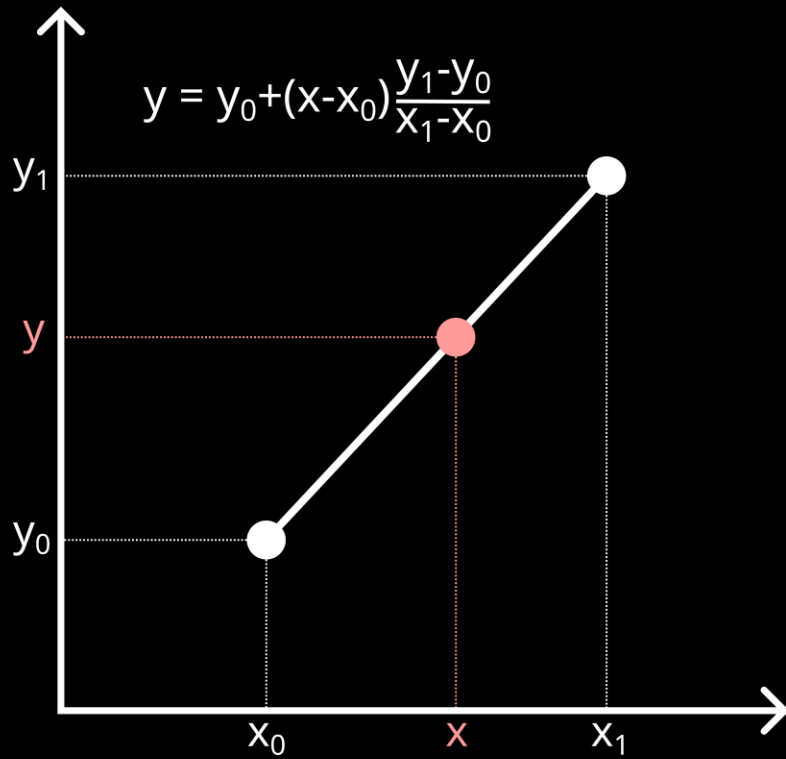
Deep learning approach



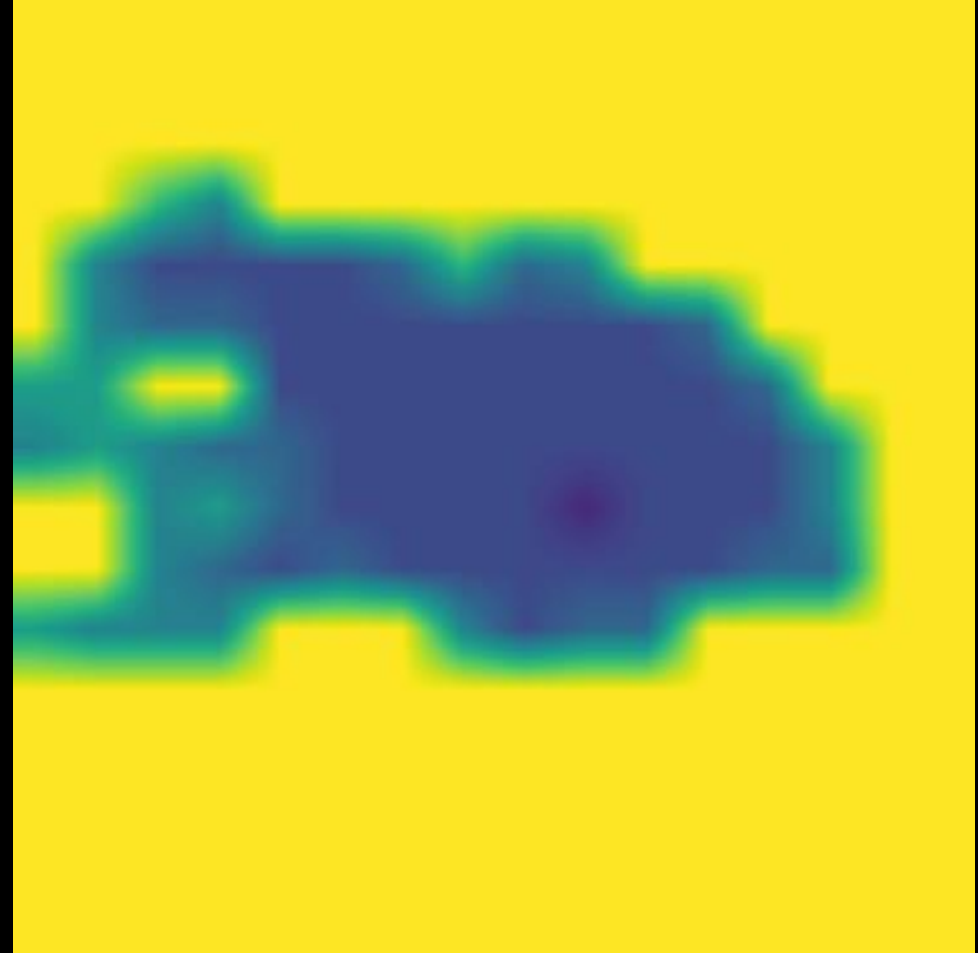
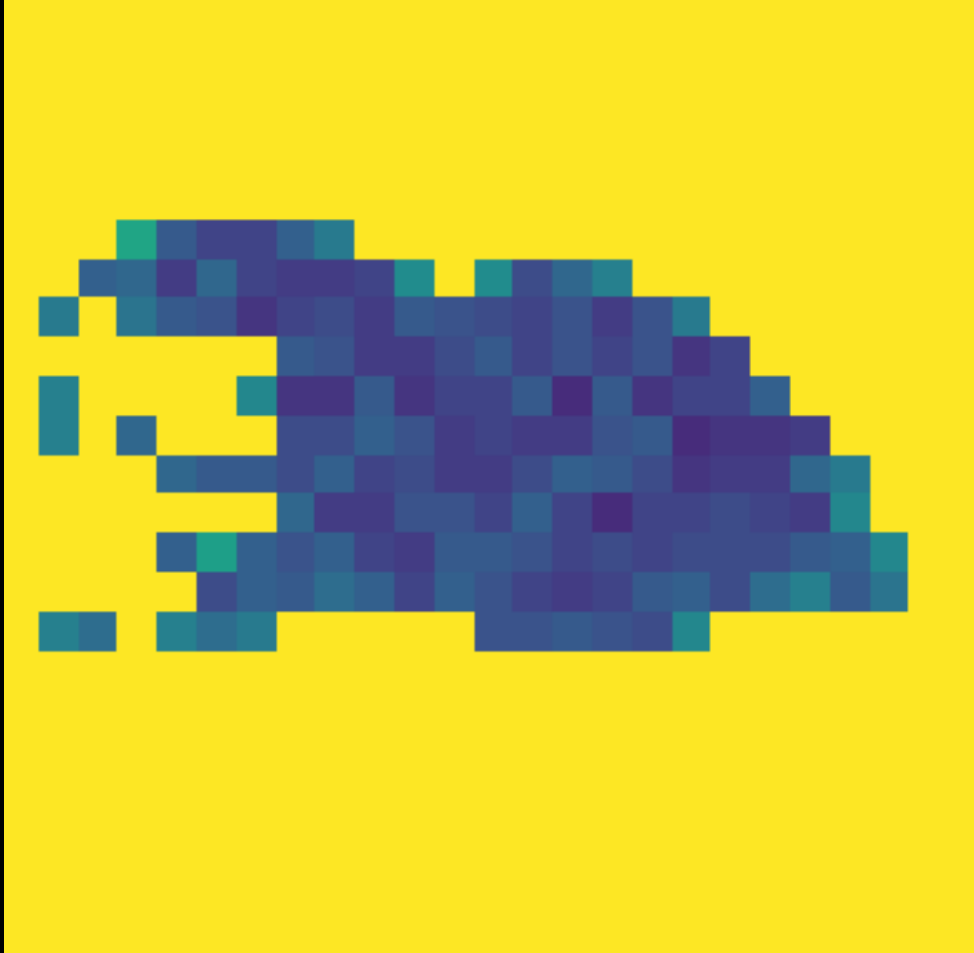
MIRROR PADDING



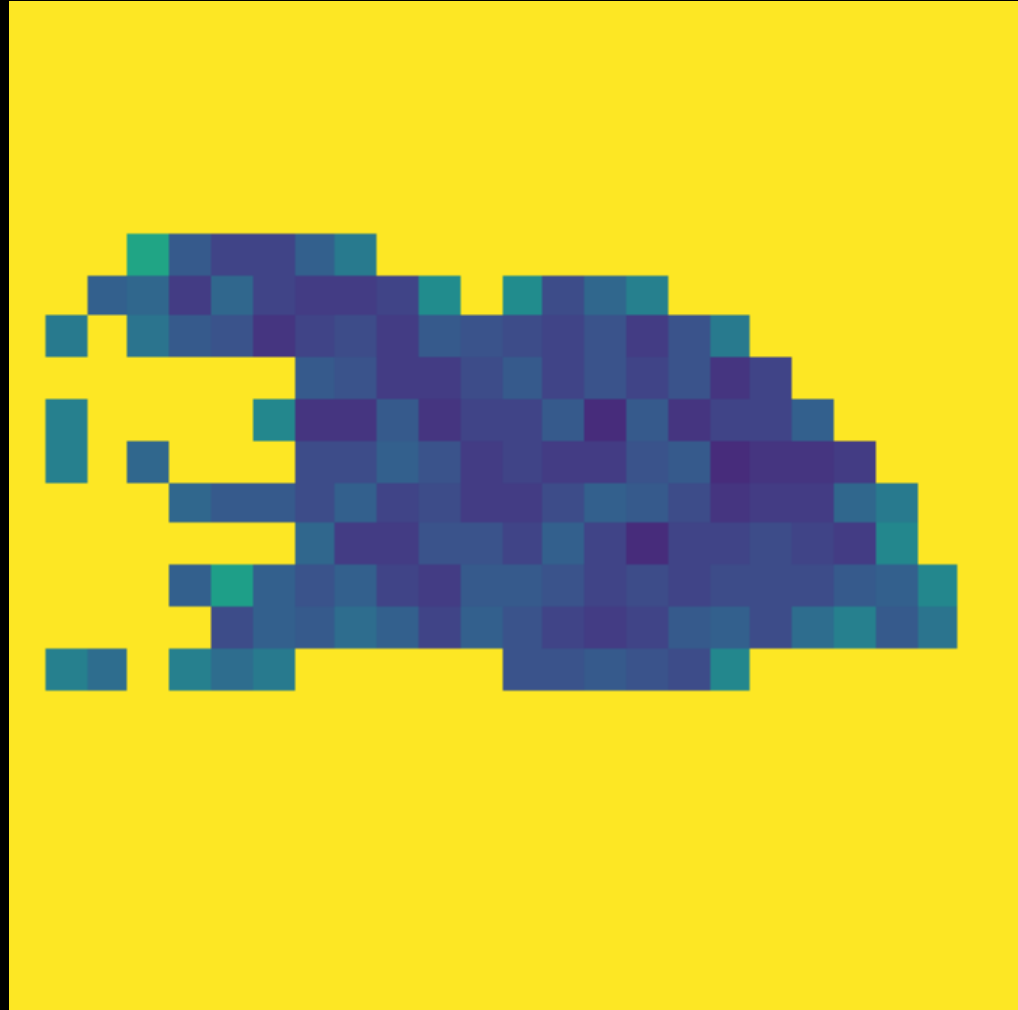
INTERPOLATION



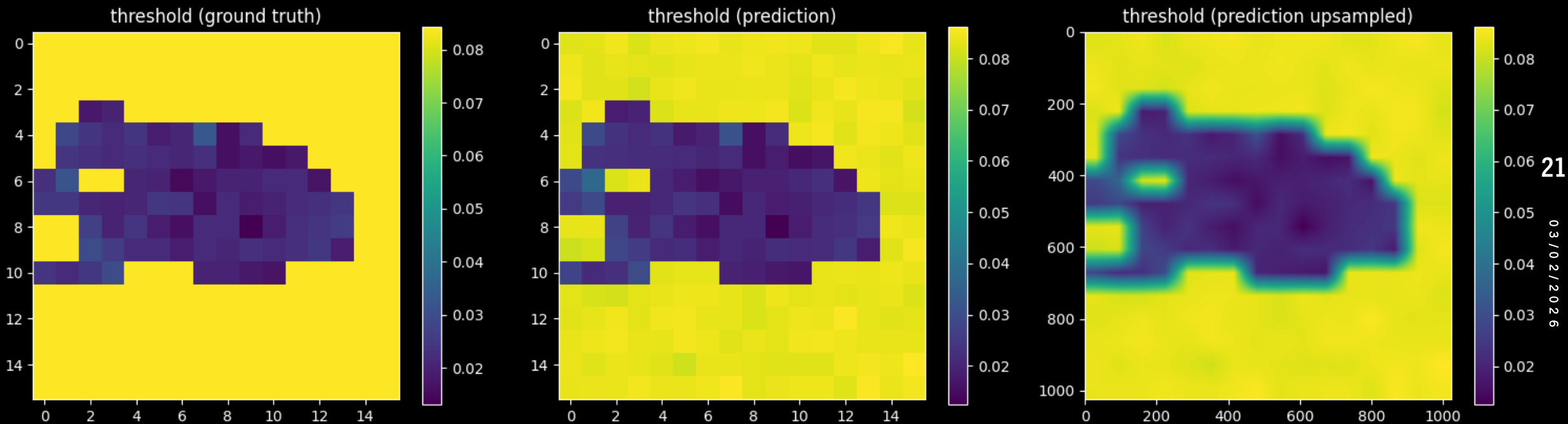
INTERPOLATION



RESULTS OF THE GROUND TRUTH

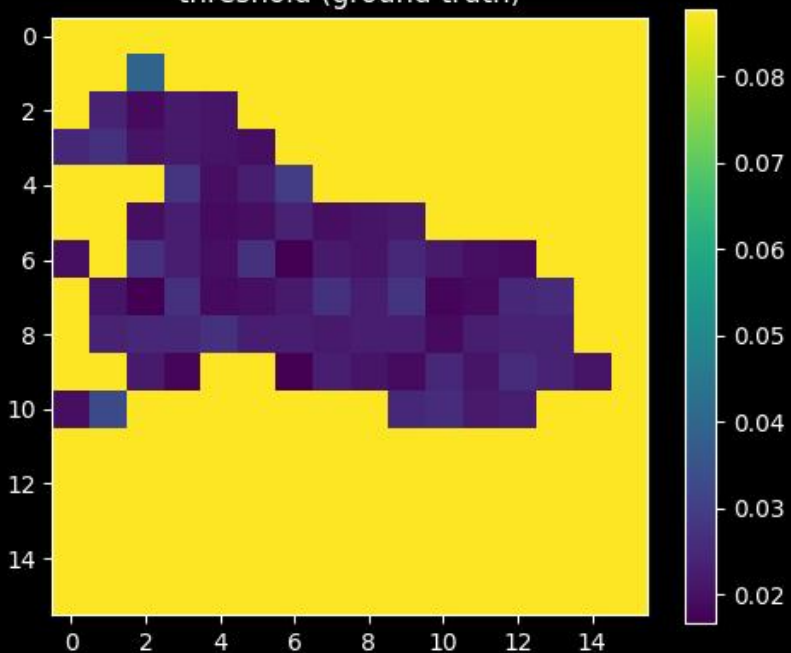


RESULTS OF THE PREDICTION

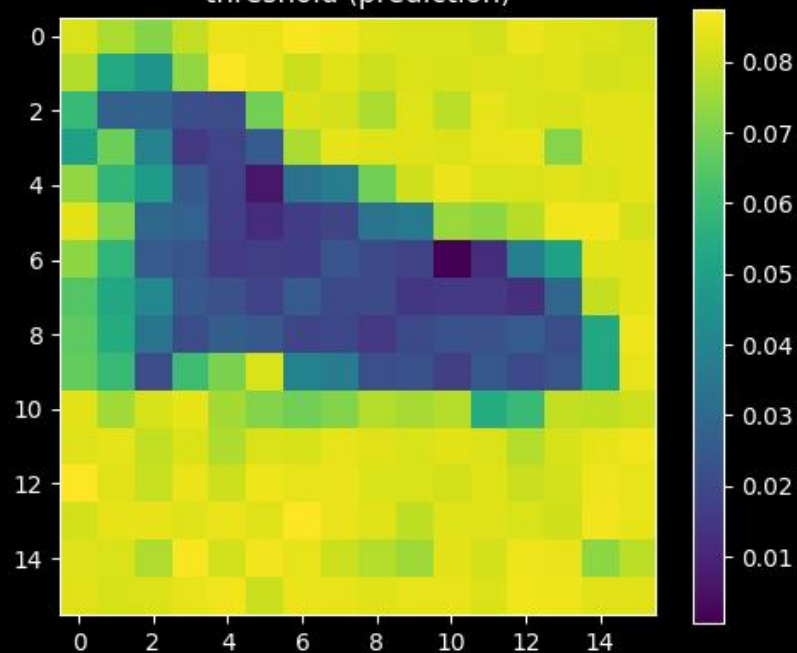


RESULTS OF THE PREDICTION

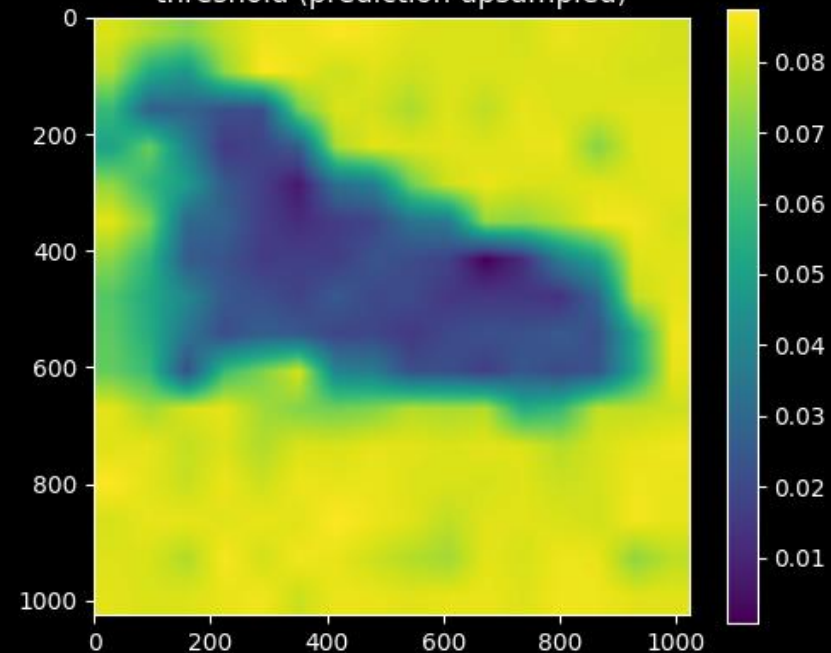
threshold (ground truth)



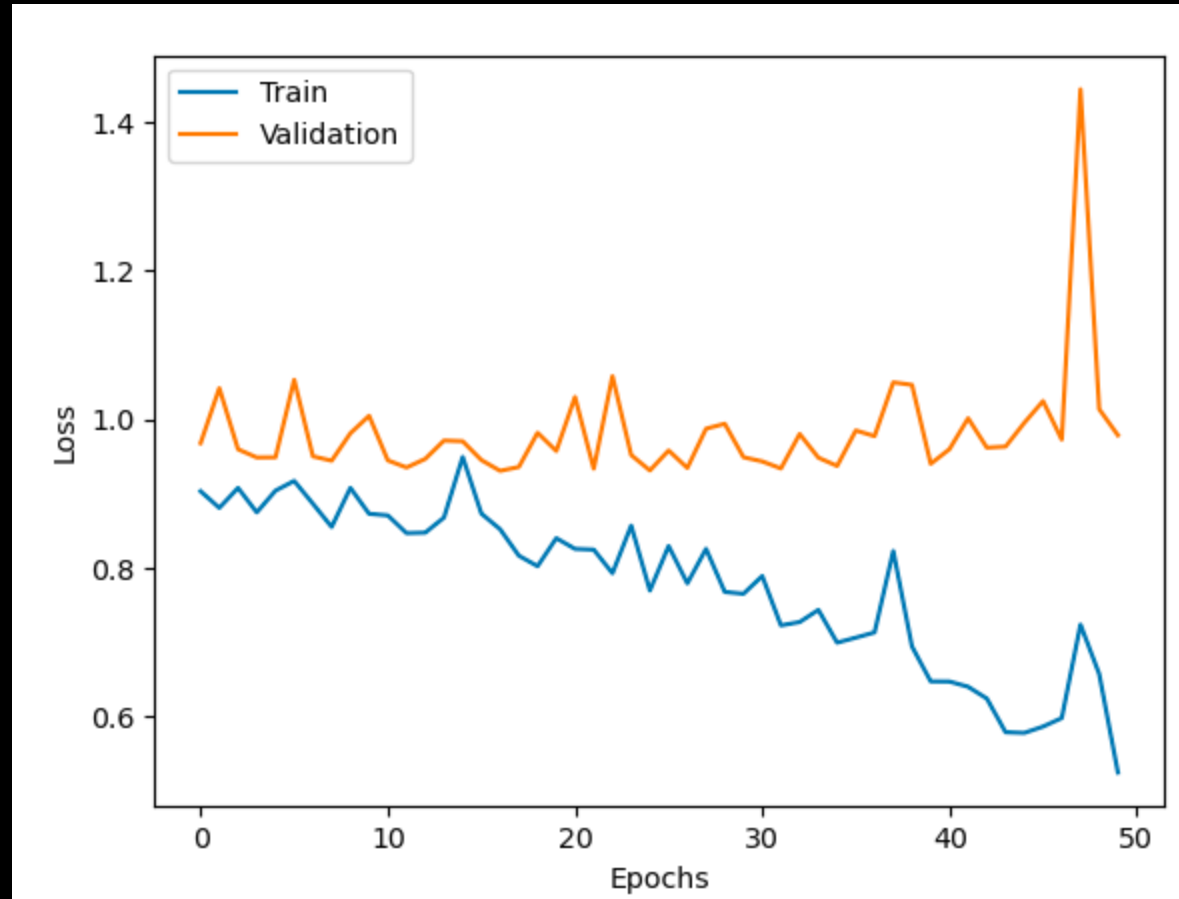
threshold (prediction)



threshold (prediction upsampled)



NEURAL NETWORK: OVERFITTING



RESULTS ON THE SIMULATOR

SANS MOD F1		SANS MOD IOU	
method	mean_score	method	mean_score
bf_f1_patch_global_std	0.940467219291635	bf_iou_patch_global_std	0.666140568857248
bf_f1_patch_global_std	0.935849586603774	bf_iou_patch_global_std	0.663365669630827
bf_f1_patch_global_std	0.914811972371451	bf_iou_patch_global_std_interp	0.655951580363147
std_local_clamp_interp	0.912844036697248	bf_iou_patch_global_std	0.655531943153141
bf_f1_patch_global_std	0.911877394638015	bf_iou_patch_global_std_interp	0.654607074771595
bf_f1_patch_global_std_interp	0.910921766072812	std_global	0.652666021167908
bf_f1_patch_global_std_interp	0.908242612752722	bf_iou_patch_global_std	0.652460613935358
bf_f1_patch_global_std_interp	0.905246671887236	std_global	0.651344397819126
std_local_clamp	0.905087319665907	bf_iou_patch_global_std_interp	0.651262916948008
bf_iou_patch_global_std	0.901149425287356	bf_iou_patch_global_std_interp	0.642242850317385
bf_f1_patch_global_std_interp	0.900548159749413	std_local_clamp_interp	0.636090548224439
std_global	0.897865853658537	std_local_clamp	0.630957492486046
std_global	0.897338403041825	bf_iou_patch_global_std_interp	0.61702686724036
bf_iou_patch_global_std_interp	0.895545314900154	bf_iou_patch_global_std_interp	0.61702686724036
bf_iou_patch_global_std	0.895545314900154	bf_iou_patch_global_std_interp	0.611244174003107
bf_iou_patch_global_std_interp	0.89247311827857	bf_iou_patch_global_std_interp	0.611244174003107
bf_iou_patch_global_std_interp	0.892307692307692	bf_f1_patch_global_std_interp	0.601730001478634
bf_iou_patch_global_std	0.888379204892966	bf_f1_patch_global_std	0.600043742937338
bf_iou_patch_global_std_interp	0.888379204892966	bf_f1_patch_global_std_interp	0.586717647932289
bf_f1_patch_global_std	0.883610451306413	bf_f1_patch_global_std	0.584758432087511
bf_f1_patch_global_std_interp	0.883610451306413	bf_f1_patch_global_std	0.579184360814175
bf_iou_patch_global_std_interp	0.882303276385725	bf_f1_patch_global_std	0.57514564268135
bf_f1_patch_global_std_interp	0.877893056664006	bf_f1_patch_global_std_interp	0.57514564268135
bf_f1_patch_global_std_interp	0.877893056664006	bf_f1_patch_global_std_interp	0.573839978993961
bf_iou_patch_global_std_interp	0.860681114551084	bf_f1_patch_global_std	0.5619639869784
bf_iou_patch_global_std_interp	0.860681114551084	bf_f1_patch_global_std	0.559676544786645
bf_iou_patch_global_std_interp	0.846390168970814	bf_f1_patch_global_std_interp	0.559676544786645
bf_iou_patch_global_std_interp	0.846390168970814	bf_f1_patch_global_std_interp	0.546709129511677
std_local_without_clamp	0.521739130434783	std_local_without_clamp	0.262818310461711
AVEC MOD F1		AVEC MOD IOU	
method	mean_score	method	mean_score
bf_f1_patch_global_std_interp	0.91086004691165	bf_iou_patch_global_std_interp	0.648526794349082
bf_f1_patch_global_std_interp	0.909923664122137	bf_iou_patch_global_std	0.645086552146427
bf_f1_patch_global_std	0.908108108108108	bf_iou_patch_global_std_interp	0.644527817210843
bf_f1_patch_global_std	0.90656112852665	bf_iou_patch_global_std	0.643238897093142
bf_iou_patch_global_std_interp	0.896392939370683	bf_iou_patch_global_std	0.629614713364347
bf_f1_patch_global_std	0.89584964761159	std_global	0.626556976244755
bf_iou_patch_global_std_interp	0.894009216589862	bf_iou_patch_global_std_interp	0.625659778521408
std_global	0.89381207282659	bf_iou_patch_global_std	0.624538483094489
bf_f1_patch_global_std_interp	0.890995260663507	std_global	0.623523369286081
std_local_clamp_interp	0.890769230769231	bf_iou_patch_global_std_interp	0.616613090759866
bf_iou_patch_global_std	0.889661270801815	std_local_clamp_interp	0.599442976550505
bf_iou_patch_global_std	0.888888888888889	bf_iou_patch_global_std_interp	0.594105137236659
bf_iou_patch_global_std_interp	0.888379204892966	bf_iou_patch_global_std	0.594105137236659
bf_iou_patch_global_std	0.884648484848485	std_local_clamp	0.593851421234709
bf_iou_patch_global_std_interp	0.881901840490739	bf_iou_patch_global_std_interp	0.587733957219251
std_local_clamp	0.88078967350038	bf_iou_patch_global_std_interp	0.587733957219251
bf_iou_patch_global_std	0.879939209726444	bf_f1_patch_global_std_interp	0.577272897406146
bf_f1_patch_global_std	0.879939209726444	bf_f1_patch_global_std	0.57029918203396
std_global	0.879818594104308	bf_f1_patch_global_std	0.569932246841238
bf_f1_patch_global_std_interp	0.869983948635634	bf_f1_patch_global_std_interp	0.568438046427495
bf_f1_patch_global_std	0.864038616251006	bf_f1_patch_global_std_interp	0.543756550381794
bf_f1_patch_global_std_interp	0.864038616251006	bf_f1_patch_global_std	0.541338655431127
bf_iou_patch_global_std_interp	0.853828306264501	bf_f1_patch_global_std	0.525043113143885
bf_iou_patch_global_std_interp	0.853828306264501	bf_f1_patch_global_std	0.525043113143885
bf_iou_patch_global_std_interp	0.851851851851852	bf_f1_patch_global_std	0.51510300198661
bf_iou_patch_global_std	0.851851851851852	bf_f1_patch_global_std_interp	0.503226780390233
bf_f1_patch_global_std_interp	0.84828712561175	bf_f1_patch_global_std_interp	0.49807125028364
bf_f1_patch_global_std_interp	0.84828712561175	bf_f1_patch_global_std	0.49807125028364
std_local_without_clamp	0.522522522522523	std_local_without_clamp	0.276652994257588

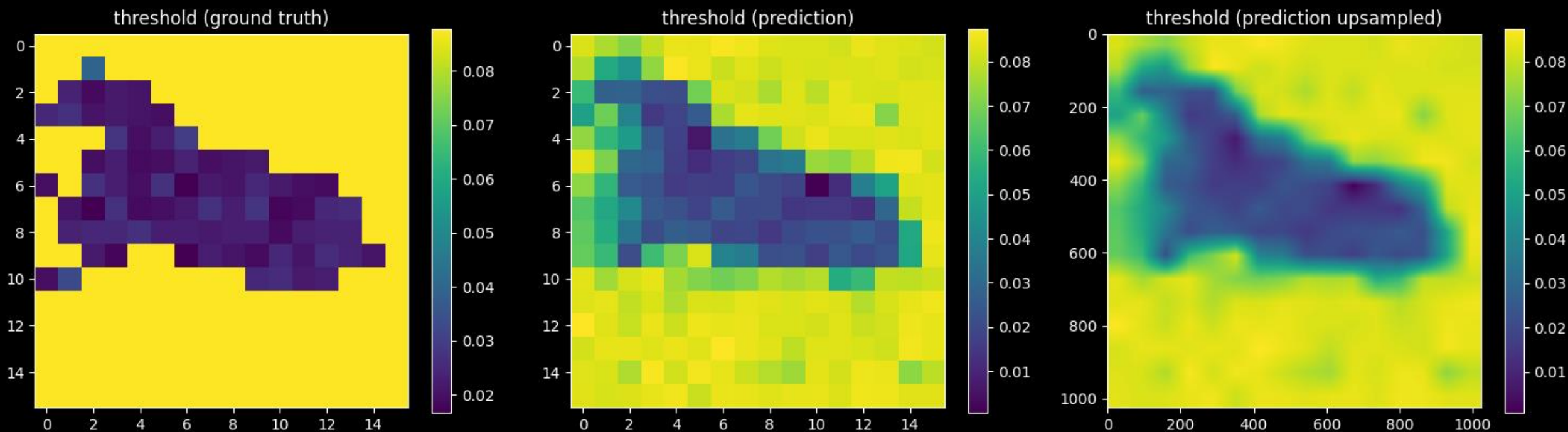
- STandard Deviation VS Median Absolute Deviation
- Interpolation VS No interpolation
- Raw images VS Modified images (heterogeneous noise)
- Brute force F₁-score VS Brute force IoU

RESULTS ON THE SIMULATOR

	Optimal results	Global threshold	Local threshold	Neural network
Jaccard Index	0.663	0.651	0.631	0.628
F -score	0.940	0.897	0.905	0.885

- Conditions :
- Standard deviation
 - No interpolation
 - No modification
 - Brute force F_1 -score for F_1 -score metric
 - Brute force IoU for IoU metric
 - Average of 7 independent network trainings (50 epochs)

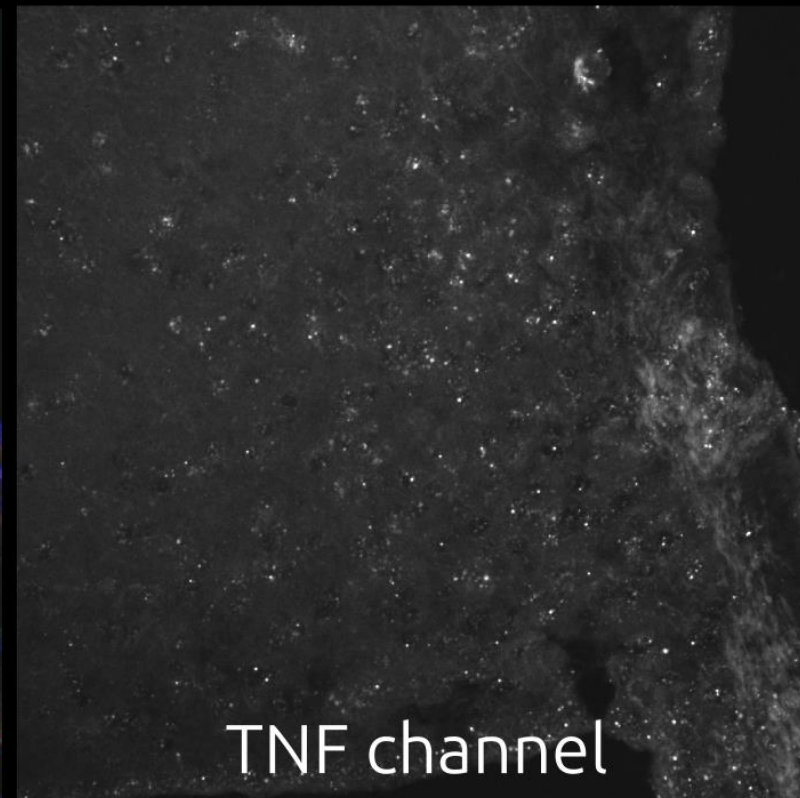
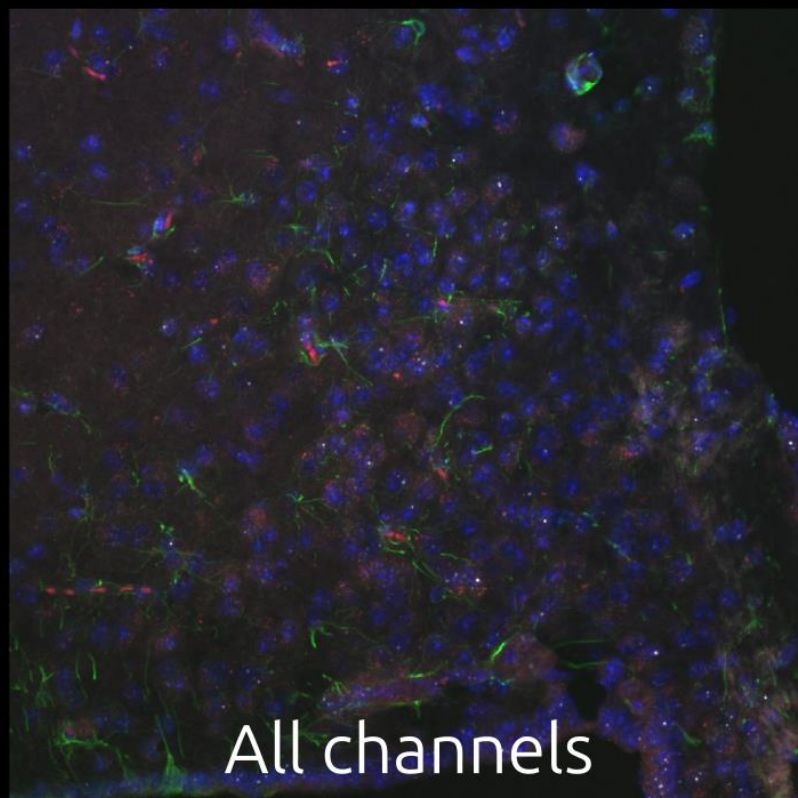
DUAL USE OF THE THRESHOLD MAP



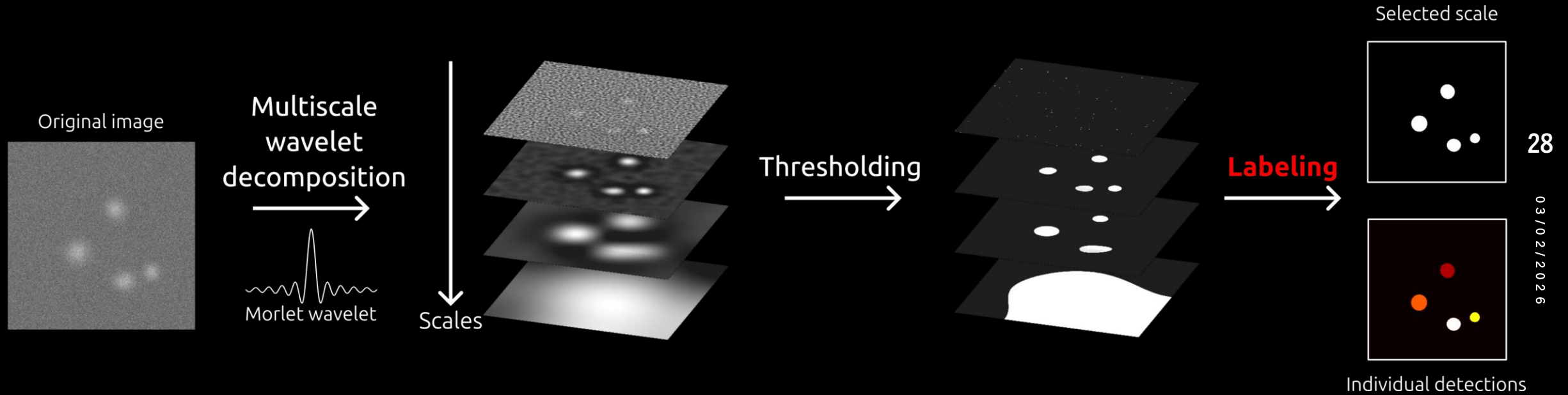
10 0

Values of k

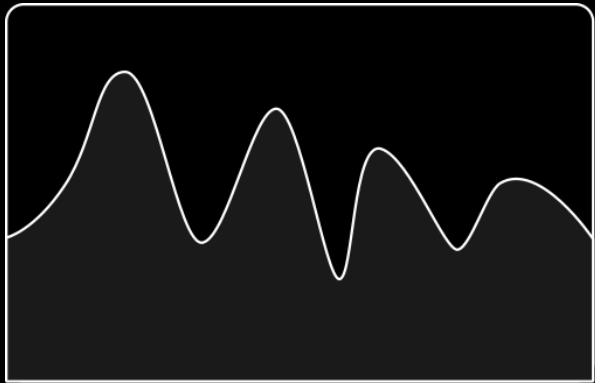
APPLICATION ON REAL IMAGES



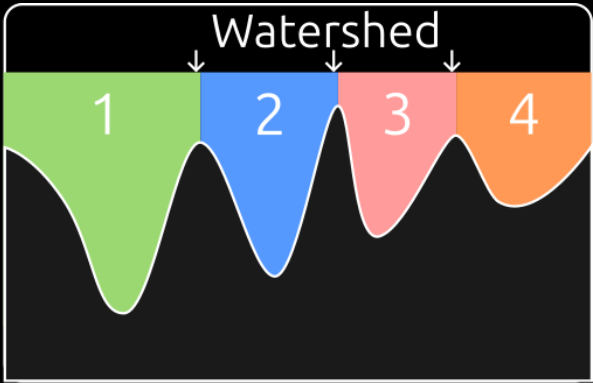
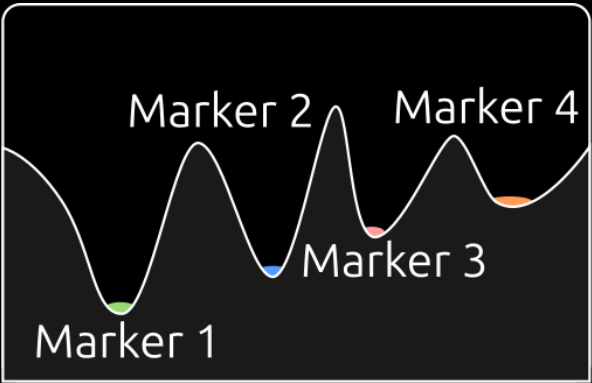
WHERE CAN THE DETECTOR BE IMPROVED?



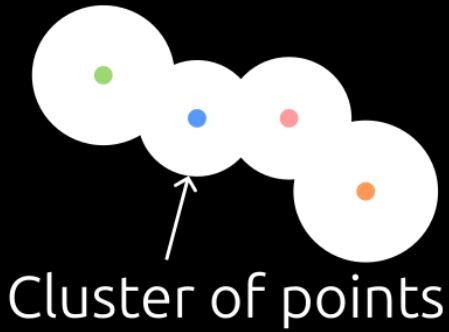
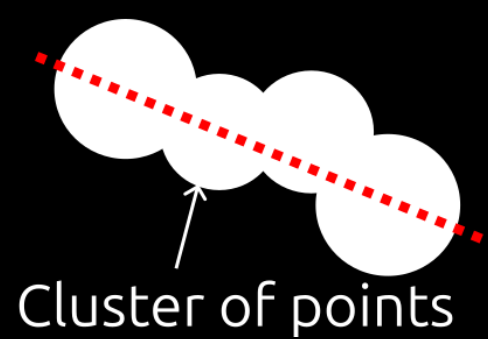
WATERSHED



Rotation



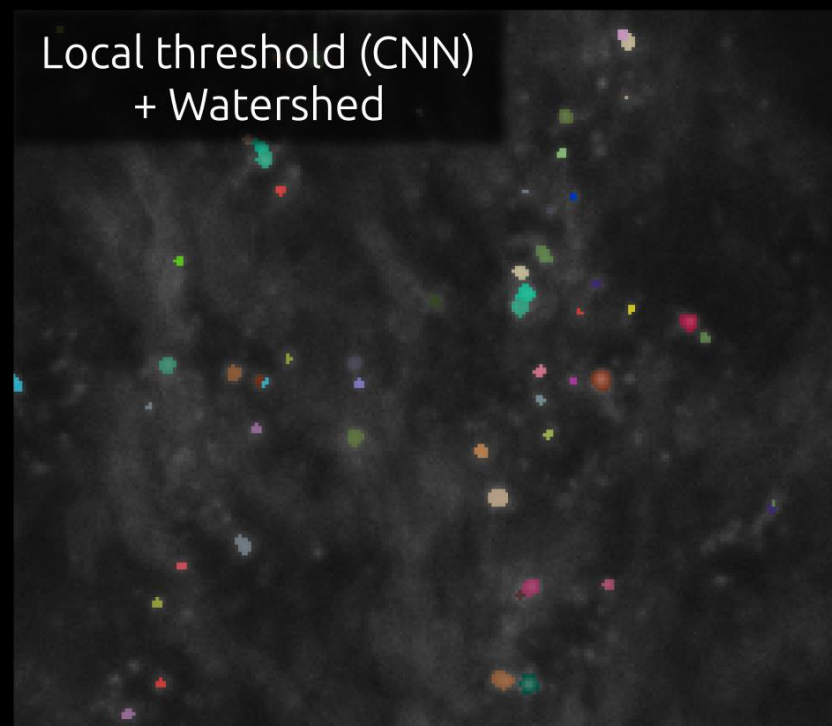
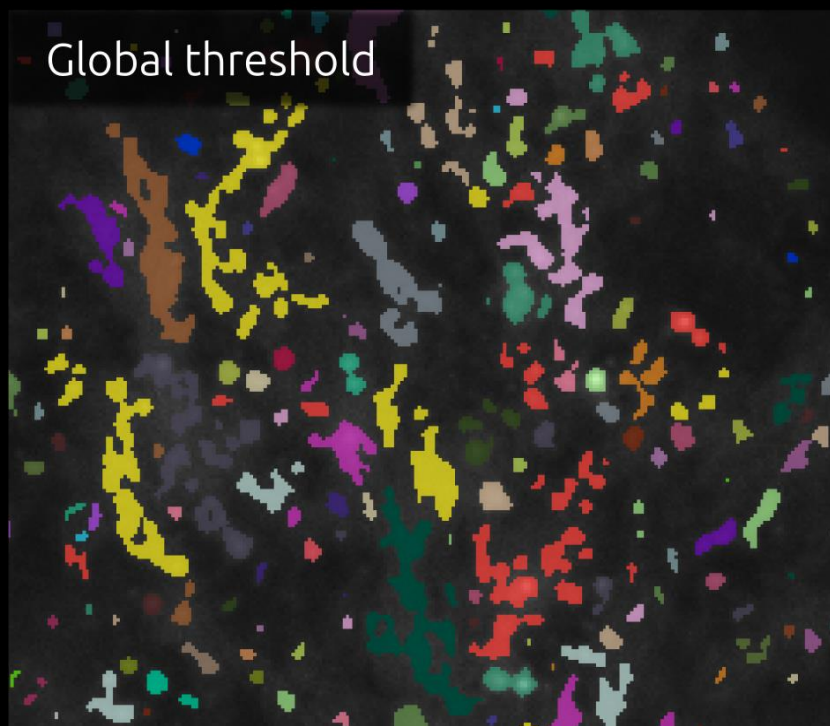
- Distance transform
- Wavelet coefficients
- Original pixels



Watershed



APPLICATION ON REAL IMAGES



THANK YOU FOR YOUR ATTENTION

